

Guided Notes: Making k-Nearest Neighbors Predictions — Complete

These notes focus on what k-nearest neighbors (k-NN) does at prediction time. The central procedure is simple: measure distance from a query point to the training data, keep the k closest observations, and aggregate their outcomes.

Learning goals

By the end of these notes, you should be able to calculate distances from a query point to training observations, identify the k nearest neighbors, make unweighted and distance-weighted k-NN predictions, and explain how modeling choices affect a prediction.

1. The k-NN prediction recipe

Let the training data be

$$D = \{(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)\},$$

where $x_i \in \mathbb{R}^p$ is the feature vector for observation i , and y_i is its outcome. For a new query point x_0 , k-NN follows this procedure:

1. Choose a distance function $d(x_0, x_i)$.
2. Compute the distance from x_0 to every training point.
3. Keep the k training points with the smallest distances; call this set $N_k(x_0)$.
4. Aggregate their outcomes to produce $\hat{y}(x_0)$.

x_0	The new query point whose outcome we want to predict.
x_i	The feature vector for training observation i .
$d(x_0, x_i)$	The distance from the query to training point i .
$N_k(x_0)$	The set of the k training observations closest to x_0 .
k	The number of neighbors used in the prediction.

Why is a distance function part of the k-NN model, rather than merely an implementation detail?

The distance function determines which observations are neighbors. Changing it can change the neighborhood and therefore the prediction.

2. Distances determine the neighborhood

For feature vectors $x = (x_1, \dots, x_p)$ and $z = (z_1, \dots, z_p)$, common choices include

Euclidean (L_2)	$d_2(x, z) = \sqrt{\sum_{j=1}^p (x_j - z_j)^2}$	Numeric features with comparable scales
Manhattan (L_1)	$d_1(x, z) = \sum_{j=1}^p x_j - z_j $	Axis-aligned differences; less emphasis on large gaps
Hamming	$d_H(x, z) = \sum_j \mathbf{1}\{x_j \neq z_j\}$	Binary, categorical, or fixed-length discrete features
Mahalanobis	$d_M(x, z) = \frac{1}{\sqrt{(x - z)^\top \Sigma^{-1} (x - z)}}$	Correlated numeric features with a reliable covariance estimate

Let $x_0 = (2, 1)$ and $x_i = (5, 5)$. Then

$$d_2(x_0, x_i) = \sqrt{(2 - 5)^2 + (1 - 5)^2} = \sqrt{25} = 5,$$

$$d_1(x_0, x_i) = |2 - 5| + |1 - 5| = 3 + 4 = 7.$$

Will Euclidean and Manhattan distance always rank two candidate neighbors in the same order? Explain.

No. The two metrics combine coordinate-wise differences differently, so two candidate points can be ordered differently under L_1 and L_2 distance.

Counterexample. Let the query be $x_0 = (0, 0)$, with candidates $A = (5, 0)$ and $B = (3, 3)$. Under Manhattan distance, $d_1(x_0, A) = 5$ and $d_1(x_0, B) = 6$, so A is closer. Under Euclidean distance, $d_2(x_0, A) = 5$ while $d_2(x_0, B) = \sqrt{18} \approx 4.24$, so B is closer.

Scale is part of the distance calculation

Suppose one feature is age (roughly 18–80) and another is annual income (roughly \$20,000–\$200,000). Before applying L_1 or L_2 distance, what should you consider doing? Why?

Standardize, robust-scale, or otherwise put the features on a comparable meaningful scale. Without scaling, income differences will usually dominate the distance solely because they are measured in much larger units.

3. Constructing the neighborhood

For a query x_0 , the k -nearest neighborhood contains the k points with the smallest distances:

$$N_k(x_0) = \{(x_i, y_i) : x_i \text{ is among the } k \text{ closest training points to } x_0\}.$$

If $x_i \in N_k(x_0)$ and $x_j \notin N_k(x_0)$, then $d(x_0, x_i) \leq d(x_0, x_j)$.

Use Euclidean distance and the query point

$$x_0 = (2, 1).$$

A	(2, 2)	Red	8	1	1 (tied)
B	(3, 1)	Blue	10	1	1 (tied)
C	(0, 1)	Red	4	2	3
D	(2, 4)	Blue	14	3	4

$$N_3(x_0) = \{A, B, C\}.$$

The radius of the smallest Euclidean ball around x_0 containing these three neighbors is 2, the distance to C.

Boundaries

Suppose the second- and third-smallest distances are equal, but $k = 2$. What decision must an implementation make?

It must use a documented tie rule: for example, choose exactly k points using a deterministic secondary rule, choose randomly with a fixed seed, or include all observations tied at the boundary and allow the effective neighborhood to exceed k .

4. Unweighted k-NN predictions

Classification: majority vote

For a class label c , count how many of the neighbors have that label:

$$\hat{y}(x_0) = \arg \max_c \sum_{i \in N_k(x_0)} \mathbf{1}\{y_i = c\}.$$

Red	2
Blue	1

$$\hat{y}_{\text{class}}(x_0) = \text{Red}.$$

A reasonable tie-breaking rule is to select the class of the closest tied neighbor. Other valid rules include random selection with a fixed seed or a fixed class order.

Regression: uniform average

For a numeric outcome, unweighted k-NN uses the arithmetic mean:

$$\hat{y}_{\text{reg}}(x_0) = \frac{1}{k} \sum_{i \in N_k(x_0)} y_i = \frac{8 + 10 + 4}{3} = \frac{22}{3} \approx 7.33.$$

With $k = 1$, the prediction depends on the tie rule because A and B are equally close. It would be 8 if A is chosen and 10 if B is chosen. A small k is highly local and can have lower bias, but it has higher variance and is more sensitive to noise and ties.

5. Distance-weighted k-NN predictions

Uniform k-NN gives each selected neighbor the same influence. A distance-weighted version gives closer neighbors more influence. One common choice is inverse-distance weighting:

$$w_i = \frac{1}{d(x_0, x_i)}.$$

For regression, the weighted prediction is

$$\hat{y}_{\text{wreg}}(x_0) = \frac{\sum_{i \in N_k(x_0)} w_i y_i}{\sum_{i \in N_k(x_0)} w_i}.$$

For classification, one weighted-vote version is

$$\hat{y}_{\text{wclass}}(x_0) = \arg \max_c \sum_{i \in N_k(x_0)} w_i \mathbf{1}\{y_i = c\}.$$

Use the $k = 3$ neighborhood from Section 3 and $w_i = 1/d_2(x_0, x_i)$.

A	1	8	1	8	Red
B	1	10	1	10	Blue
C	2	4	$\frac{1}{2}$	2	Red
Total			$\sum w_i = \frac{5}{2}$	$\sum w_i y_i = 20$	

The weighted regression prediction is

$$\hat{y}_{\text{wreg}}(x_0) = \frac{20}{5/2} = 8.$$

The weighted class scores are

$$\text{score}(\text{Red}) = 1 + \frac{1}{2} = \frac{3}{2}, \quad \text{score}(\text{Blue}) = 1.$$

$$\hat{y}_{\text{wclass}}(x_0) = \text{Red}.$$

The uniform regression prediction is about **7.33**, whereas the weighted prediction is 8. Neighbor C has the smallest response (4) but is farther away, so its influence is reduced by weighting.

The zero-distance case

If $d(x_0, x_i) = 0$, then $1/d(x_0, x_i)$ is undefined. Do not use the inverse-distance formula directly. A sensible rule is to predict using only exact matches: average their numeric responses for regression or take their majority class for classification. If there is one exact match, use its outcome.

6. Choosing k and the prediction rule

Small k	Very local and responsive to nearby observations	Lower bias, higher variance
Large k	Smoother and less sensitive to a single neighbor	Higher bias, lower variance
Distance weighting	Closer points have more influence within the chosen neighborhood	Can preserve locality while using several neighbors; may be sensitive to extremely small distances

Why should preprocessing, distance metric, weighting rule, and k be selected together using validation data?

They jointly determine the neighbors and their influence, so each can change predictions and out-of-sample performance. Validation estimates the performance of the complete pipeline rather than tuning one choice in isolation.

7. A prediction checklist

Before reporting a k -NN prediction, verify the following.

- ☐ The query point and training features use the same representation and preprocessing.
- ☐ The distance metric is appropriate for the feature types and scales.
- ☐ Distances have been computed consistently.
- ☐ Exactly k neighbors have been selected, with a documented rule for boundary ties.
- ☐ The aggregation rule matches the outcome type: vote for classification, average for regression.
- ☐ A zero-distance rule has been specified when using inverse-distance weights.

Exit questions

1. Can changing the distance metric change a k -NN prediction even when k is fixed? Why?
Yes. A different metric can rank the training observations differently, which changes the neighbor set and can change the aggregated prediction.
2. In weighted k -NN regression, why does the denominator $\sum_{i \in N_k(x_0)} w_i$ appear?
It normalizes the weights so the prediction is a weighted average. The result remains on the outcome scale and is unchanged if all weights are multiplied by the same positive constant.
3. A classifier has neighbors with labels A, A, B, B for $k = 4$. Give two possible tie-breaking rules.
For example: choose the class of the closest neighbor among the tied classes, or choose randomly with a fixed seed. A fixed class order is another valid rule if documented.
4. Why can a feature measured on a very large numeric scale dominate Euclidean distance?
Euclidean distance squares and sums raw coordinate differences. A large-unit feature can contribute much more to the total distance even when it is not more important for the prediction.